- ❒ Power developed by the two stroke engine is twice that developed by the four stroke engine for the same engine speed and volume.
- ❒ Cost of the engine is less since less no. of parts
- ❒ Less maintenance since fewer spare parts due to its simple design
- ❒ Low manufacturing cost
- ❒ It has high mechanical efficiency.
- ❒ A two stroke engine is more compact, light and requires less space

Disadvantages:

- ❒ It has high fuel consumption
- ❒ It does more consumption of the lubricating oil.
- ❒ Not suitable for heavy vehicles
- ❒ There is a greater wear and tear of moving parts.
- ❒ Thermal efficiency is less than four stroke engine.
- ❒ The charge is diluted by the burnt gases due to incomplete scavenging.
- ❒ It produces greater noise.

5.6 COMPARISON OF TWO STROKE AND FOUR STROKE ENGINE

The two stroke and four stroke engine are compared and their differences are given below (Table 5.6).

Table 5.6 Differnce Between Two Stroke and Four Stroke Engine

S.no	Two Stroke Engine	Four Stroke Engine
1.	It has one revolution of crankshaft within one power stroke.	It has two revolution of crankshaft between one power strokes.
2.	2 strokes are required to complete a cycle.	4 strokes are required to complete a cycle.
3.	It requires lighter flywheel because it generates more balanced force due to one revolution for one power stroke.	It requires heavy flywheel because it generates unbalance force due to two revolutions for one power stroke.
4.	One non-power stroke in a cycle	Three non-power stroke in a cycle
5.	Engines are lighter	Engines are heavier
6.	Engine construction is simple.	The Engine construction is a bit complicated
7.	Ports are used for inlet and outlet of air fuel mixture	Valves are used for inlet and outlet of air fuel mixture
8.	Ports are opened and closed by piston movements	Valves are operated by separate valve operating mechanism.
9.	Less no. of moving parts	More no. of moving parts
10.	Volumetric efficiency is less	Volumetric efficiency is more
11.	Thermal efficiency is less	Thermal efficiency is more
12.	Two stroke engines are less efficient and generate more smoke.	Four stroke engines are more efficient and generate less smoke.

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